

The art of asking the right question - Workshop 2

Defining a Problem Statement

Overview

01. Context

02. Example use case

03. Principles for problem solving

04. Conclusion

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Why is this important

"The biggest mistake I see new programmers make is focusing on learning syntax instead of learning how to solve problems."

— V. Anton Spraul

Effective problem solving requires a **structured, transferable, logical** approach.

It can be done through critical evaluation of a problem – known as **critical thinking**.



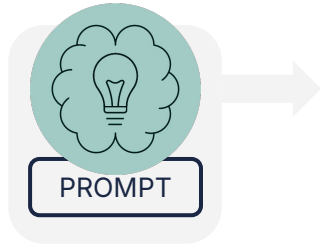
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So what is critical thinking anyway?



"Thinking about, monitoring, and regulating our own **thinking**."

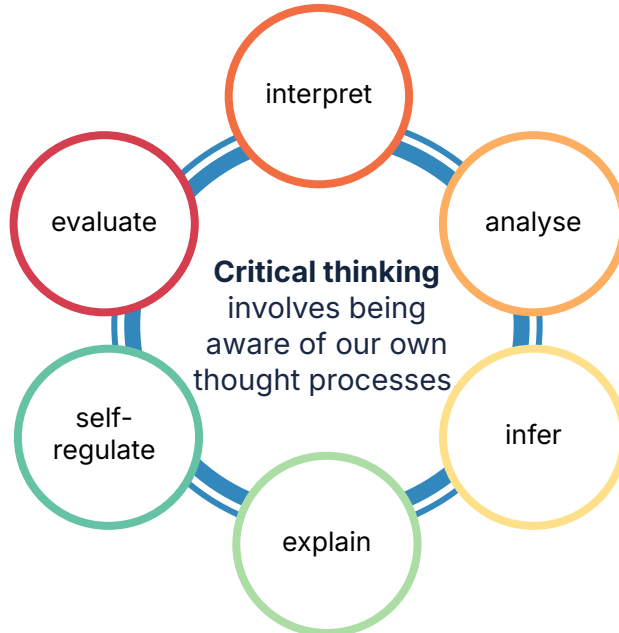
Logical & reflective thinking

Looking at your results along
the way

Awareness

Appropriate action

Range of flexible
strategies



Metacognition involves being aware of how efficient we are in applying certain strategies to complete a task.

How does it help me to solve a problem?

CRITICAL THINKING

Gives you an **intellectual toolset** that can be applied to any problem.

Ensures you **consider all parts** of the problem.

Ensures that you **correctly define** the problem.

Ensures you produce the **best solution** for a problem.



METACOGNITION

But first, try on your own



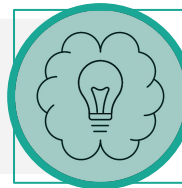
Scenario: A few of your colleagues really need coffee to get those neurons firing for a big project you're working on. They've asked you to take control of the situation.

Information: The office has a kitchen with a coffee machine and all the necessary ingredients to make coffee. You also know of a great takeaway coffee shop around the corner from the office. Make as many assumptions as you think are necessary.

Write down the steps you need to take to solve this "problem".

5
min

Try to recognise the implicit processes you follow while writing down these steps.



Let's reflect...

01. Am I looking at this project from the correct perspective?
02. Is this the best way to carry out this project?
03. Did I miss any critical information, should I re-evaluate the scope?
04. Did I follow an appropriate cognitive and logical process?
05. What are we trying to solve?

When you DON'T understand the problem



Lenny Rachitsky (former developer at Airbnb)

Build out a "social travel" experience for Airbnb travellers.

The problem:

"travellers want to hang out with other travellers."

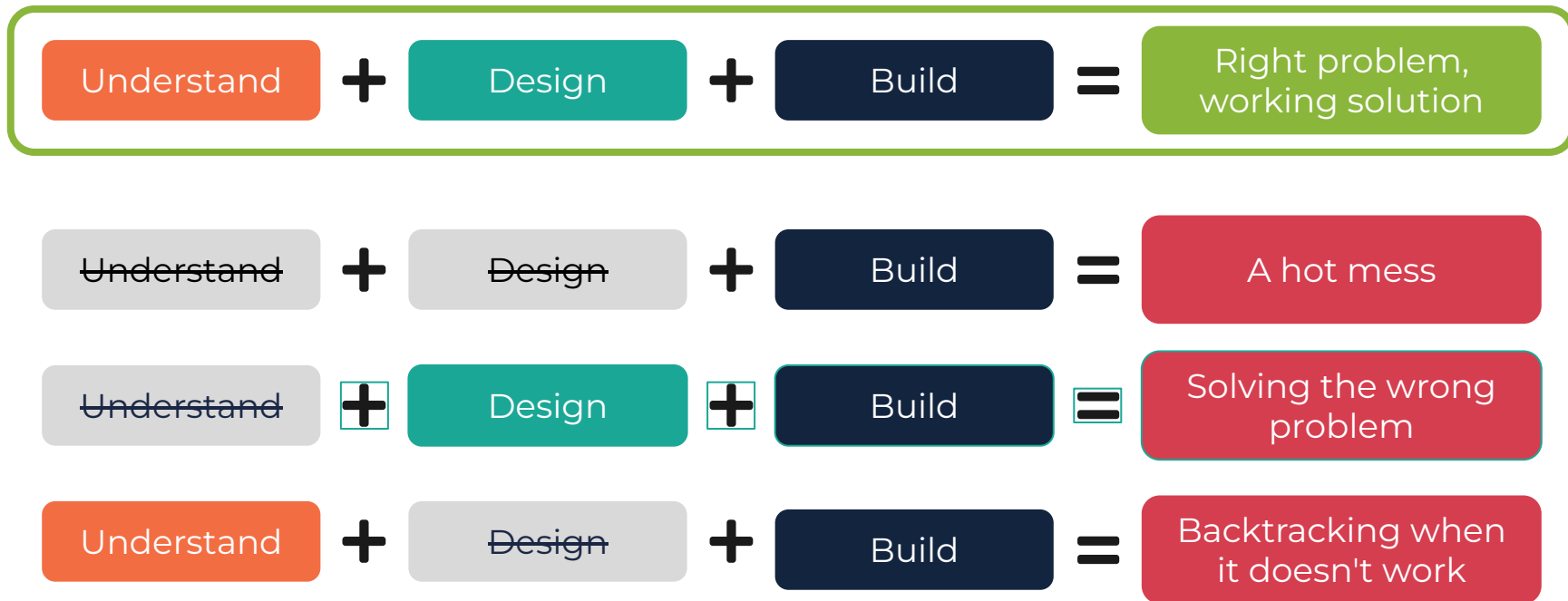
The **REAL** problem:

"travellers want to find high-quality, non-touristy things to do."

"...nothing is more certain to cause a project to fail than a misunderstanding of the problem you are solving."

A three-step approach to problem solving

Lenny's three-step process:



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Problem-solving steps to follow

UNDERSTAND THE PROBLEM

1

Understand the problem in detail. What is the desired outcome?

2

Write down a **problem statement**, making sure there is no ambiguity.

FIND THE FACTS

3

List the potential options/solutions using a **logic tree**.

4

Consider the pros and cons of each potential solution.

IMPLEMENT & REVIEW

5

Select the best option and build that solution.

Methodologies

Logic trees

A fundamental problem-structuring framework for solving complicated problems.

EGAD

A framework to guide problem solving in larger projects.

Agile

A framework to effectively manage a project. Mostly used by developers, but can be applied to almost anything.

Design Thinking

A solution-based approach to solve complex, ill-defined problems.

Scientific Method

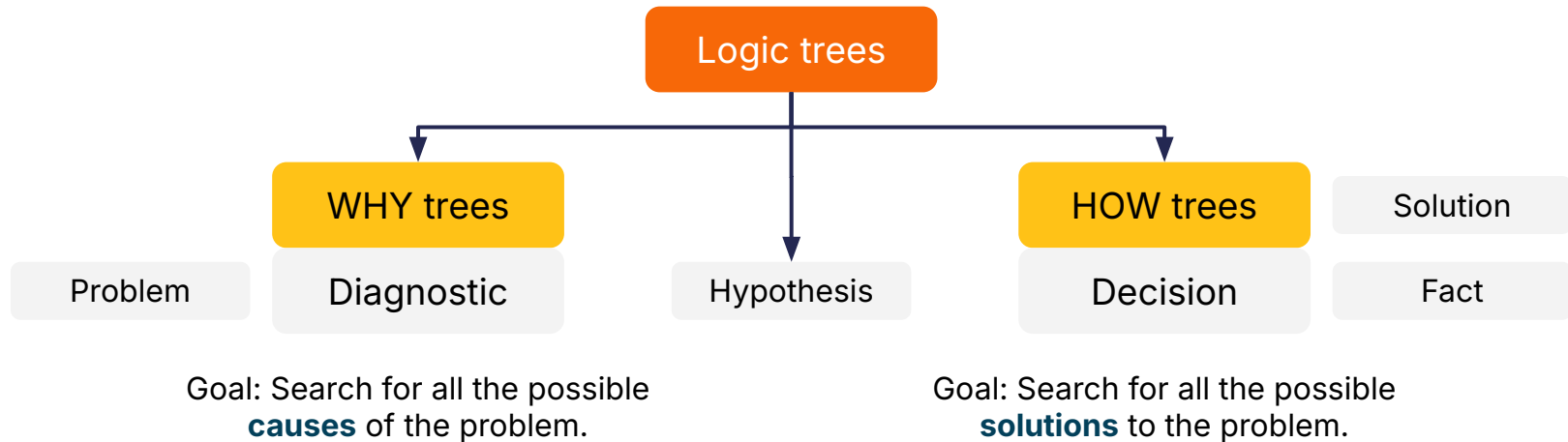
A logical problem-solving approach that is rigorous and replicable – used by scientists.

Root cause analysis

A problem-solving method used to identify the root causes of problems.

Logic Tree

A tool used to **systematically visualise** all the possible **problems, solutions, or hypotheses** in order to understand a problem.

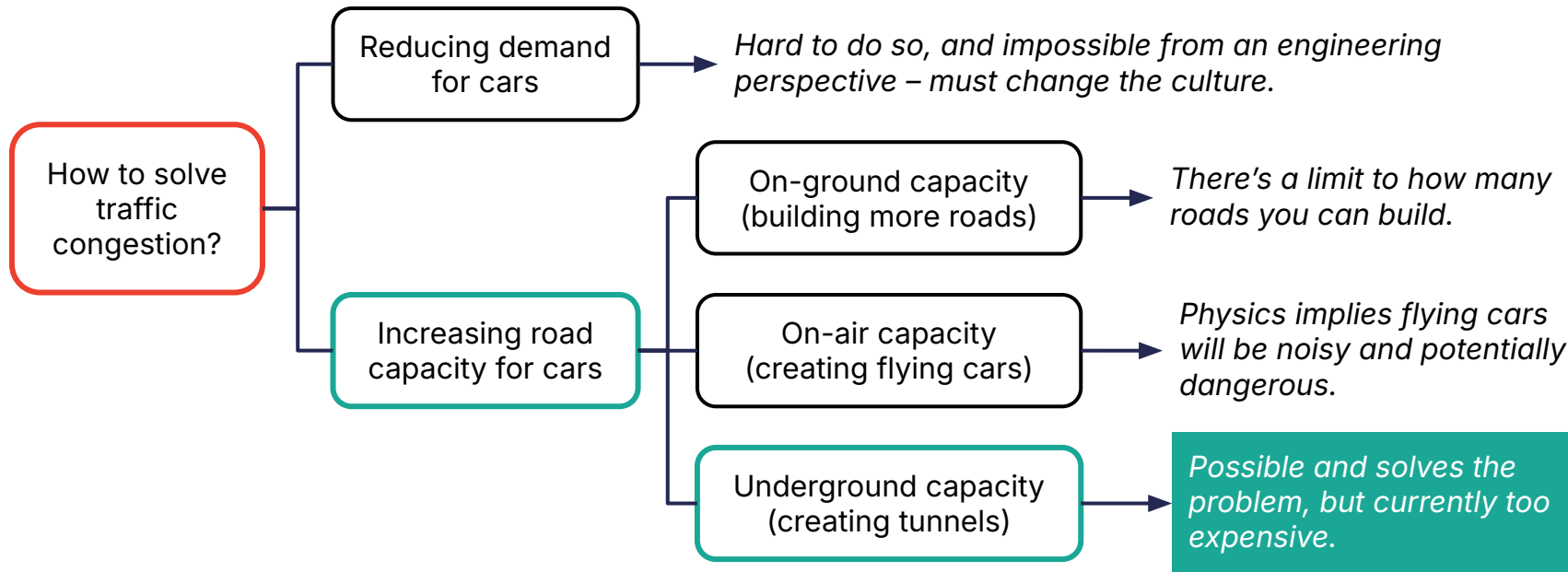


Logic Trees



How to solve the traffic congestion problem in Los Angeles

Elon's logic:



Agile

An **iterative** approach to project management and software development that assists with delivering value.

Agiles Rituals

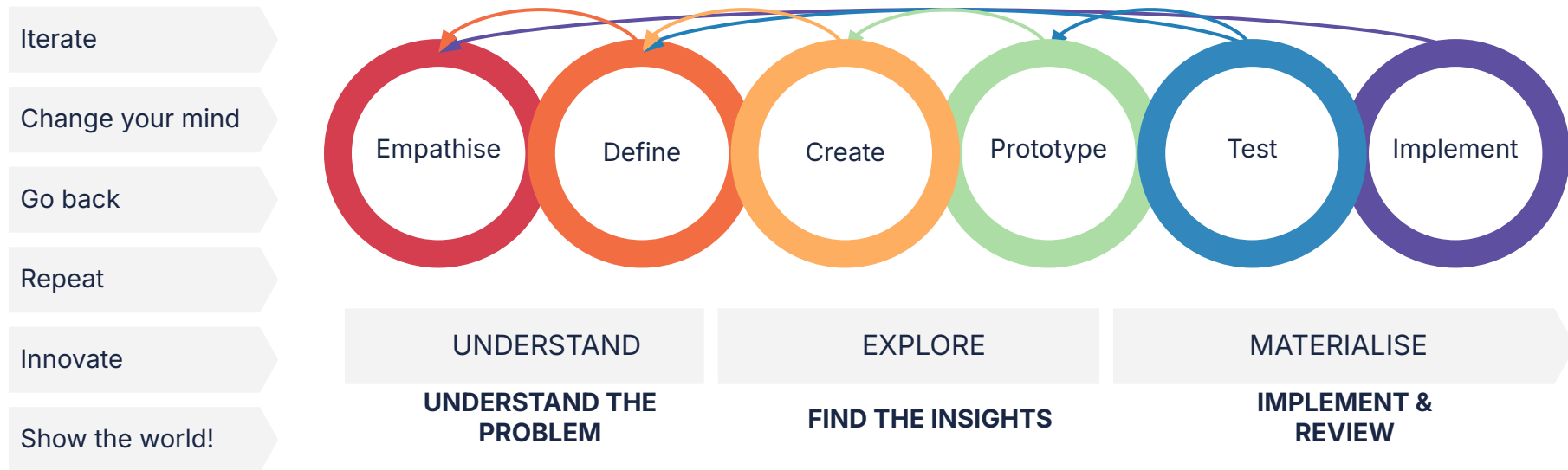
Kanban Board

User Stories



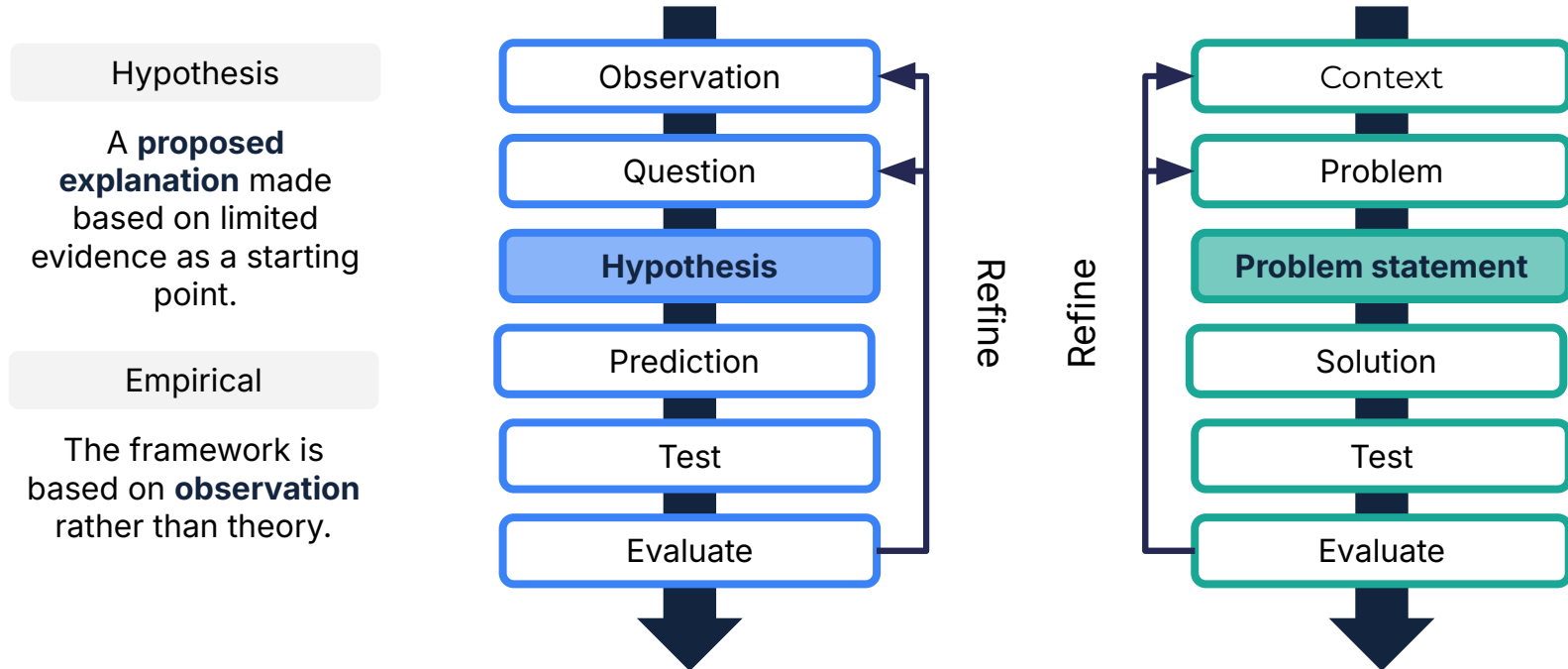
Design Thinking

An **iterative** process to **redefine** problems, **challenge** assumptions (about the problem), and create innovative solutions.



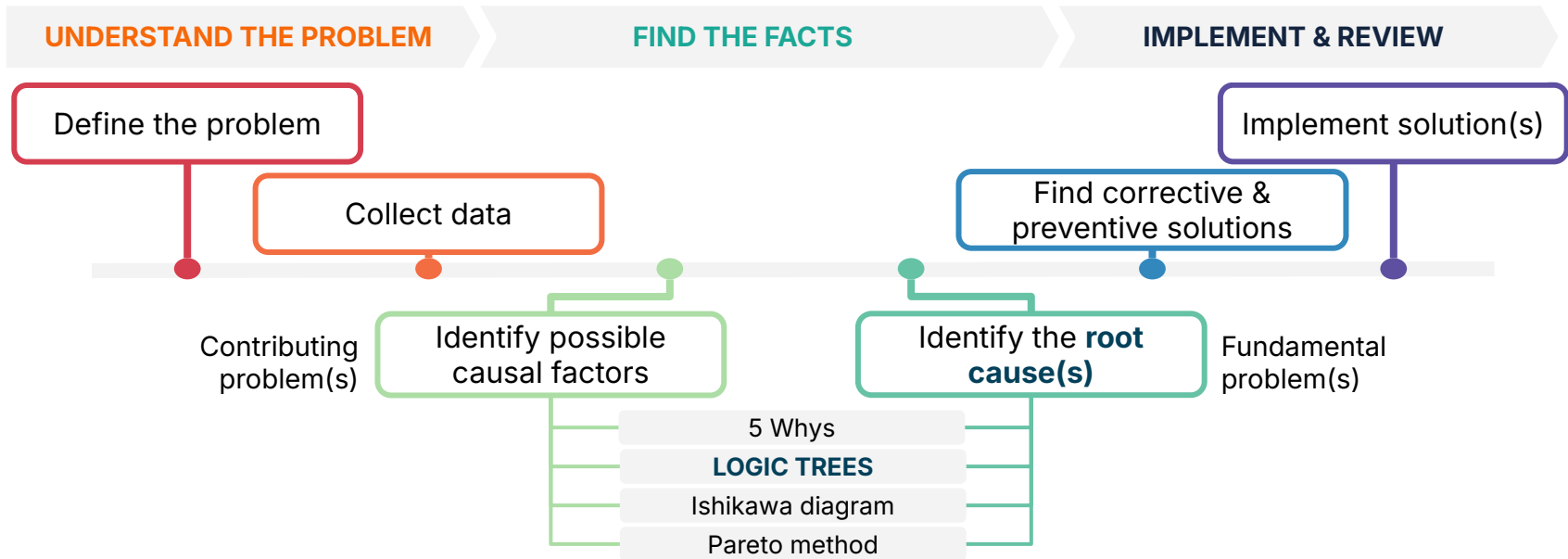
Scientific Method

Used to construct and test **hypotheses** in a **rigorous** and **replicable** manner.



Root cause analysis

A framework used to **systematically** identify and analyse the **root cause** of a problem and identify ways to resolve the problems (that lead to process failures).



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